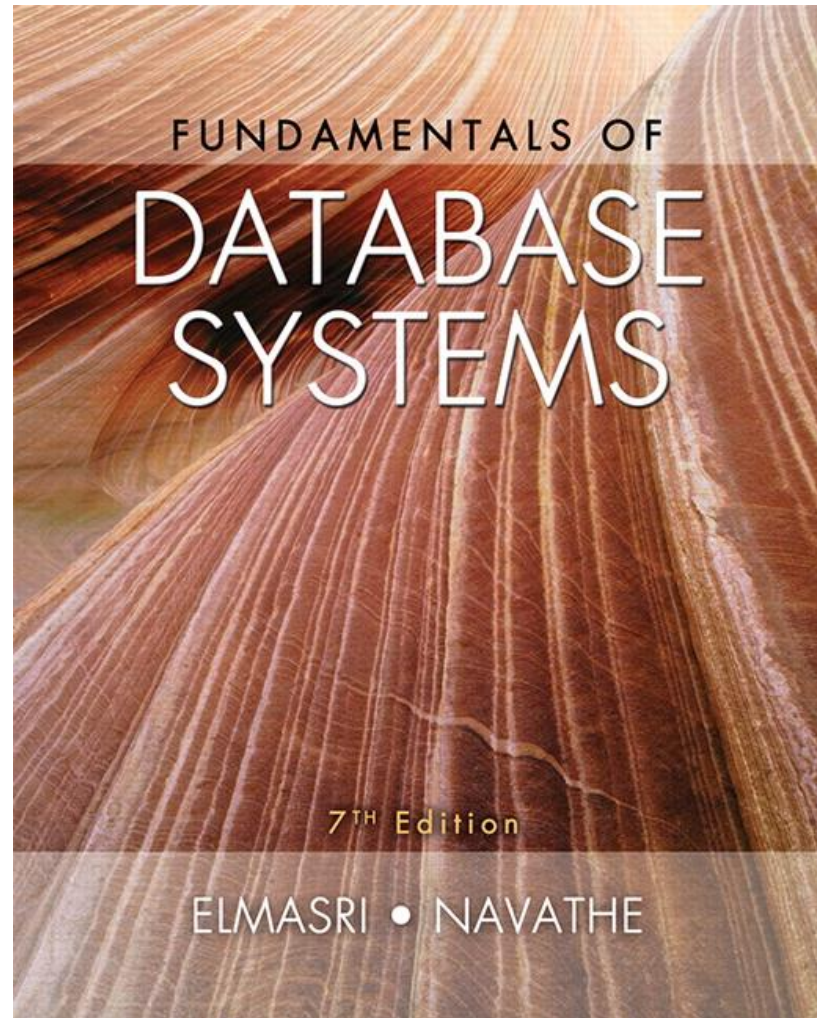




CHAPTER 19

Query Optimization

Query Processing

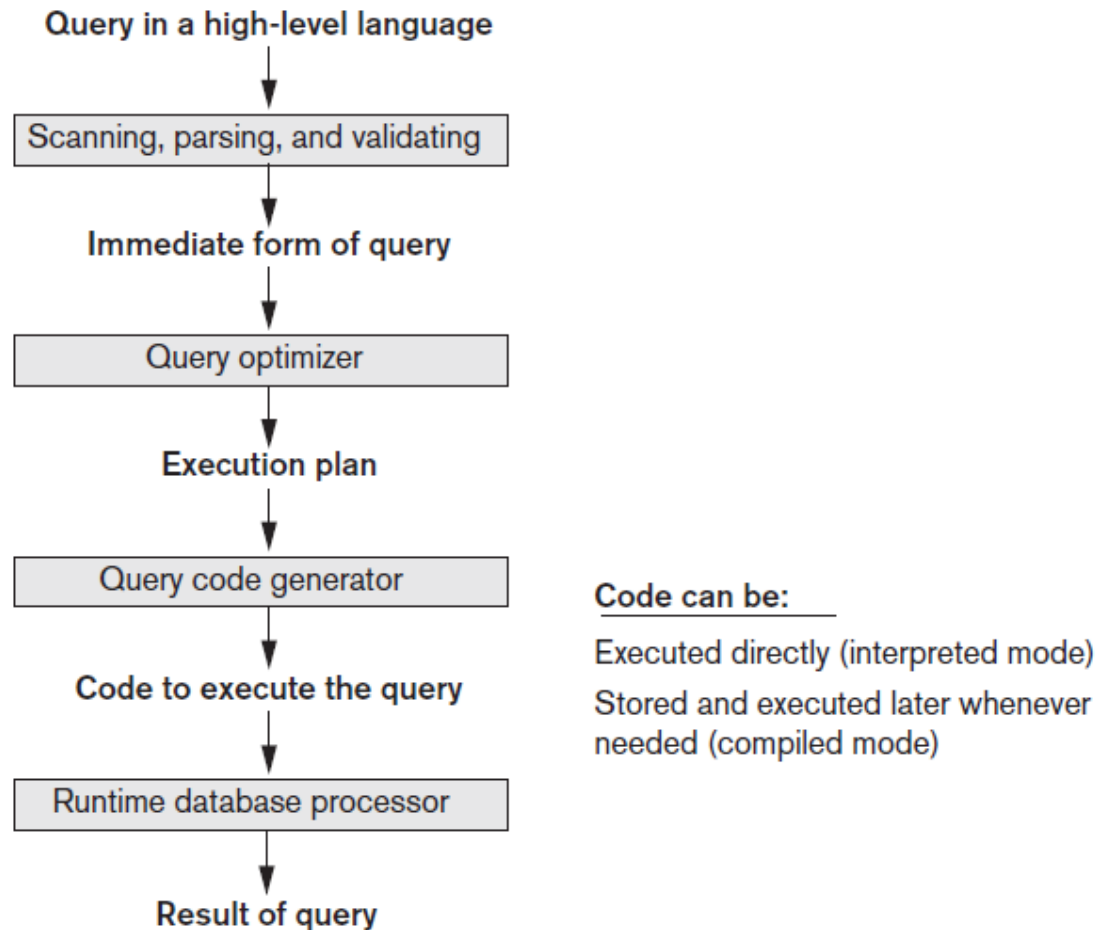


Figure 18.1 Typical steps when processing a high-level query

18.1 Translating SQL Queries into Relational Algebra and Other Operators

- SQL
 - Query language used in most RDBMSs
- Query decomposed into query blocks
 - Basic units that can be translated into the algebraic operators
 - Contains single SELECT-FROM-WHERE expression
 - May contain GROUP BY and HAVING clauses

Translating SQL Queries (cont'd.)

- Example:

```
SELECT Lname, Fname
FROM EMPLOYEE
WHERE Salary > ( SELECT MAX (Salary)
                  FROM EMPLOYEE
                  WHERE Dno=5 );
```

- Inner block

```
( SELECT MAX (Salary)
  FROM EMPLOYEE
  WHERE Dno=5 )
```

- Outer block

```
SELECT Lname, Fname
FROM EMPLOYEE
WHERE Salary > c
```

Introduction

- Query optimization
 - Conducted by a query optimizer in a DBMS
 - Goal: select best available strategy for executing query
 - Based on information available
- Most RDBMSs use a tree as the internal representation of a query

19.1 Query Trees and Heuristics for Query Optimization

- Step 1: scanner and parser generate initial query representation
- Step 2: representation is optimized according to heuristic rules
- Step 3: query execution plan is developed
 - Execute groups of operations based on access paths available and files involved

Query Trees and Heuristics for Query Optimization (cont'd.)

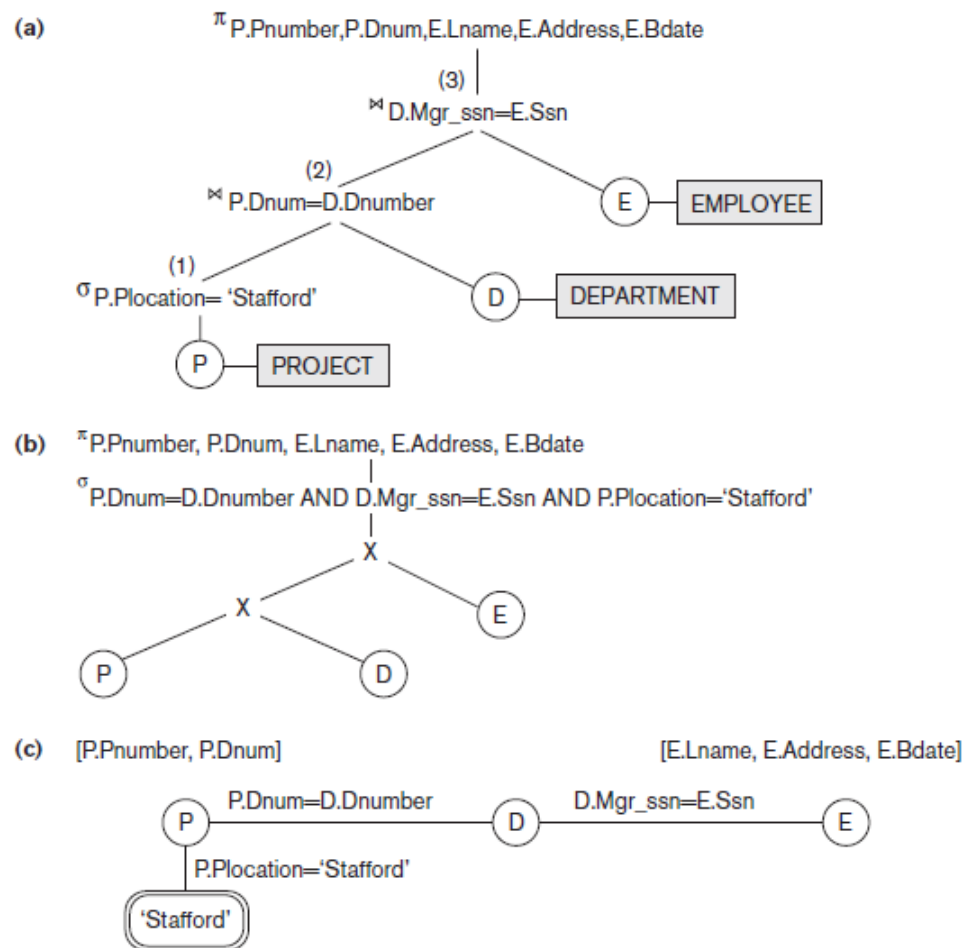
- Example heuristic rule
 - Apply SELECT and PROJECT before JOIN
 - Reduces size of files to be joined
- Query tree
 - Represents relational algebra expression
- Query graph
 - Represents relational calculus expression
- Example for Q2 on next slide

```
Q2:  SELECT  P.Pnumber, P.Dnum, E.Lname, E.Address, E.Bdate
      FROM    PROJECT P, DEPARTMENT D, EMPLOYEE E
      WHERE   P.Dnum=D.Dnumber AND D.Mgr_ssn=E.Ssn AND
              P.Plocation= 'Stafford';
```


Query Trees and Query Graph

Corresponding to Q2

Figure 19.1 Two query trees for the query Q2. (a) Query tree corresponding to the relational algebra expression for Q2. (b) Initial (canonical) query tree for SQL query Q2. (c) Query graph for Q2.



Query Trees and Heuristics for Query Optimization (cont'd.)

- Query tree represents a specific order of operations for executing a query
 - Preferred to query graph for this reason
- Query graph
 - Relation nodes displayed as single circles
 - Constants represented by constant nodes
 - Double circles or ovals
 - Selection or join conditions represented as edges
 - Attributes to be retrieved displayed in square brackets

Heuristic Optimization of Query Trees

- Many different query trees can be used to represent the query and get the same results
- Figure 19.1b shows initial tree for Q2
 - Very inefficient - will never be executed
 - Optimizer will transform into equivalent final query tree

Query Transformation Example

Q: SELECT E.Lname
 FROM EMPLOYEE E, WORKS_ON W, PROJECT P
 WHERE P.Pname='Aquarius' AND P.Pnumber=W.Pno AND E.Essn=W.Ssn
 AND E.Bdate > '1957-12-31';

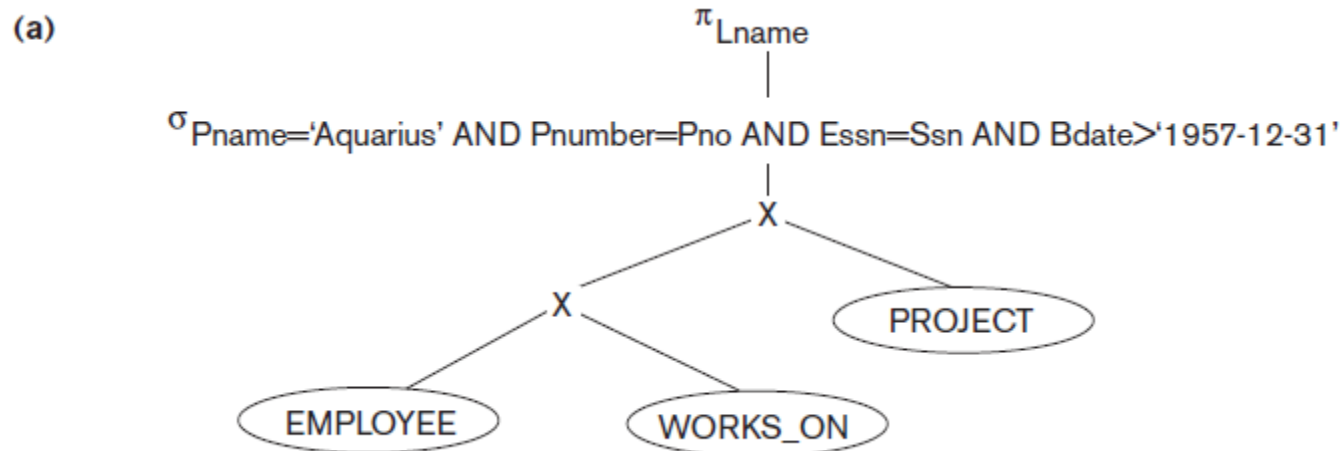


Figure 19.2 Steps in converting a query tree during heuristic optimization. (a) Initial (canonical) query tree for SQL query Q.

Query Transformation Example (cont'd.)

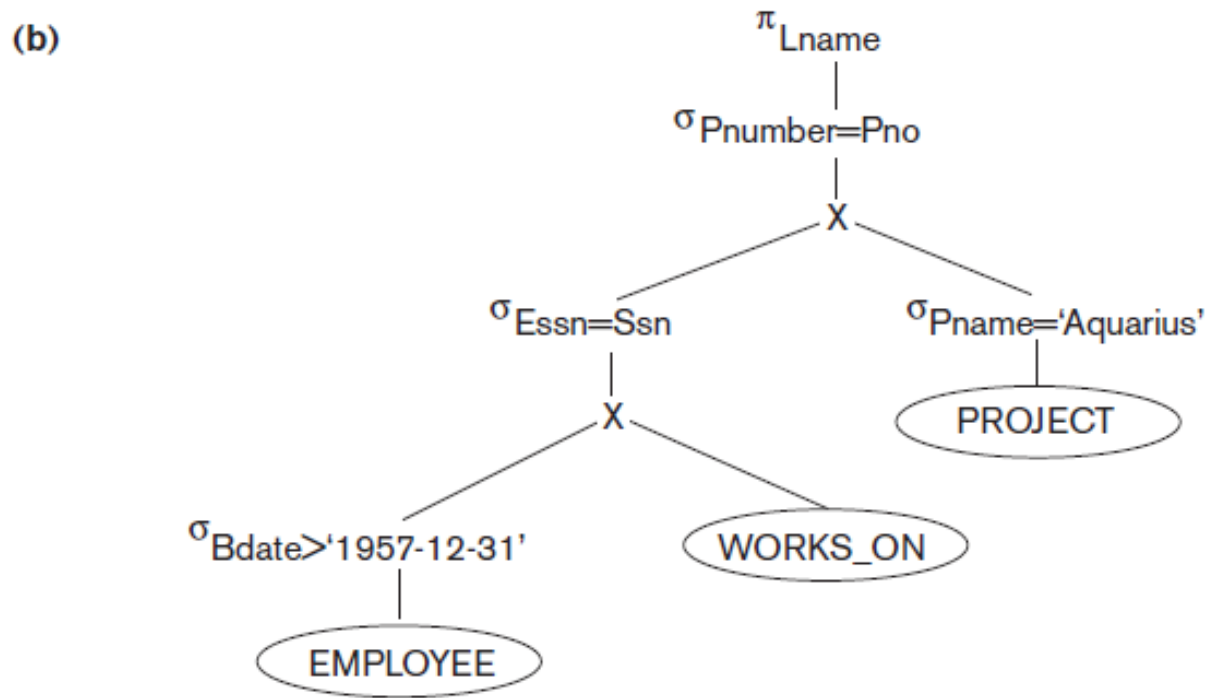


Figure 19.2 Steps in converting a query tree during heuristic optimization
(b) Moving SELECT operations down the query tree.

Query Transformation Example (cont'd.)

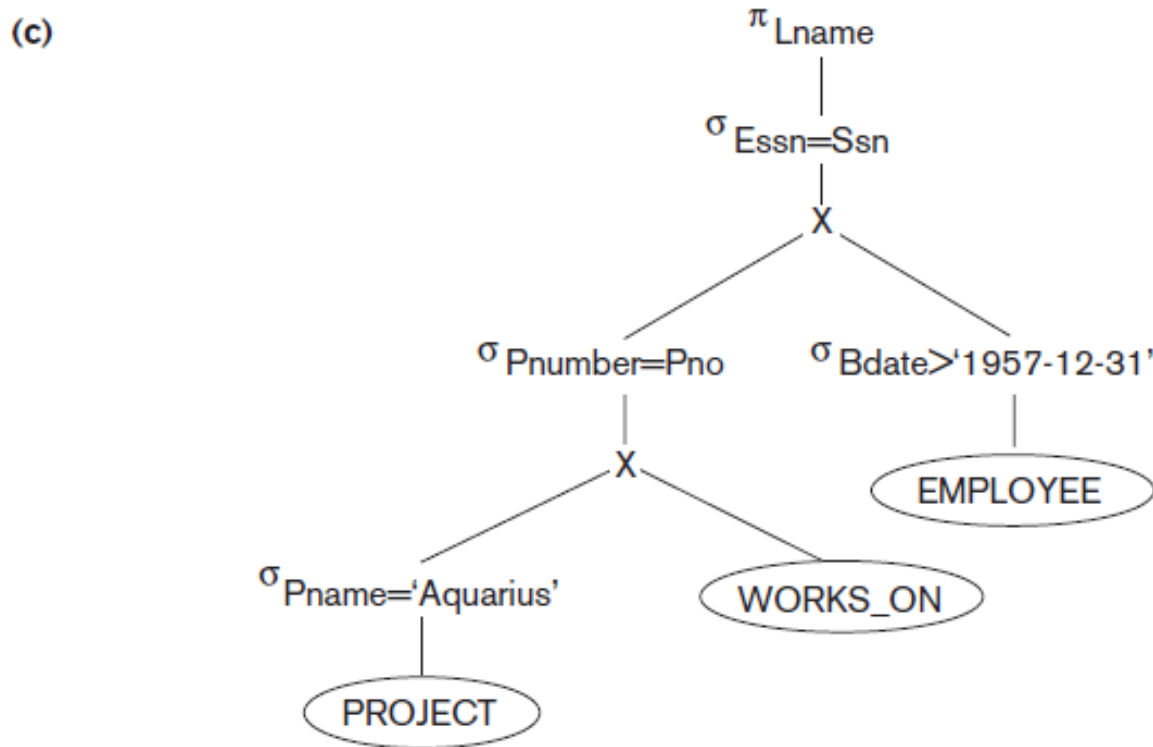


Figure 19.2 Steps in converting a query tree during heuristic optimization
(c) Applying the more restrictive SELECT operation first.

Query Transformation Example (cont'd.)

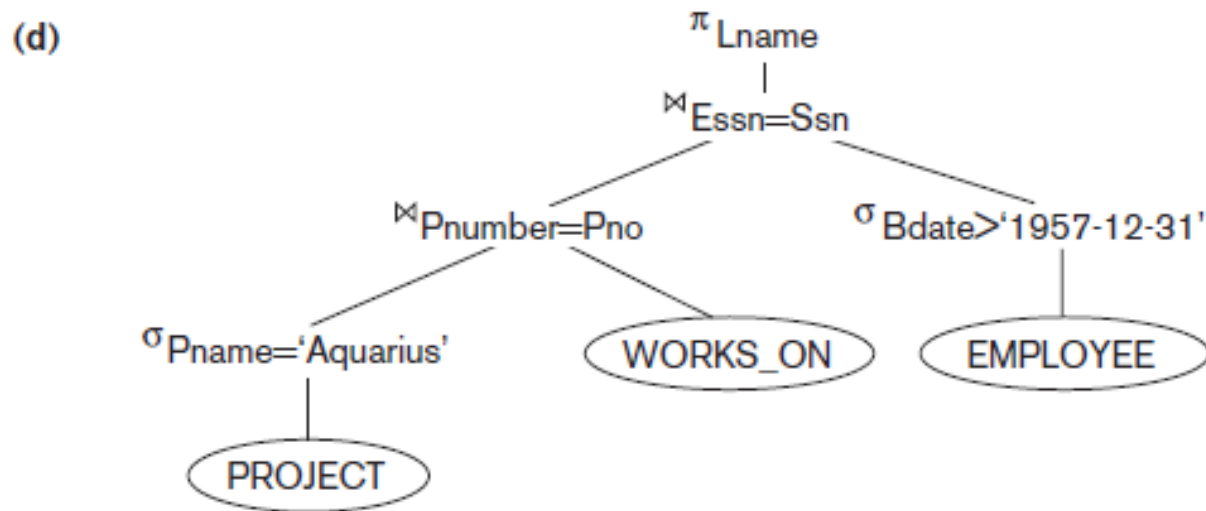


Figure 19.2 Steps in converting a query tree during heuristic optimization
(d) Replacing CARTESIAN PRODUCT and SELECT with JOIN operations.

Query Transformation Example (cont'd.)

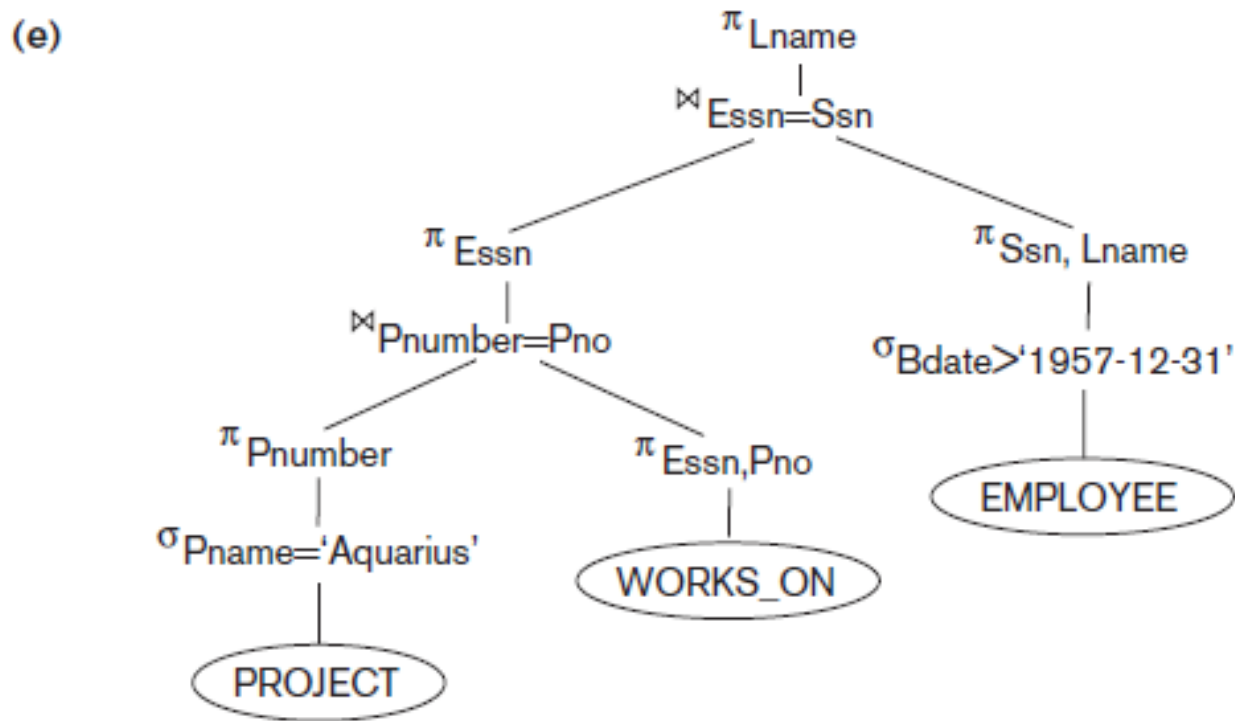


Figure 19.2 Steps in converting a query tree during heuristic optimization
(e) Moving PROJECT operations down the query tree.



Transformation of Relational Expressions

- Two relational algebra expressions are said to be **equivalent** if on every legal database instance the two expressions generate the same set of tuples
 - Note: order of tuples is irrelevant
- In SQL, inputs and outputs are multisets of tuples
 - Multiset (or bag) is a generalization of the notion of set in which members are allowed to appear more than once
 - Two expressions in the multiset version of the relational algebra are said to be equivalent if on every legal database instance the two expressions generate the same multiset of tuples
- An **equivalence rule** says that expressions of two forms are equivalent
 - Can replace expression of first form by second, or vice versa



Equivalence Rules

1. Conjunctive selection operations can be deconstructed into a sequence of individual selections.

$$s_{q_1 \cup q_2}(E) = s_{q_1}(s_{q_2}(E))$$

2. Selection operations are commutative.

$$s_{q_1}(s_{q_2}(E)) = s_{q_2}(s_{q_1}(E))$$

3. Only the last in a sequence of projection operations is needed, the others can be omitted.

$$\Pi_{L_1}(\Pi_{L_2}(\dots(\Pi_{L_n}(E))\dots)) = \Pi_{L_1}(E)$$

4. Selections can be combined with Cartesian products and theta joins.

- a. $\sigma_{\theta}(E_1 \bowtie E_2) = E_1 \bowtie_{\theta} E_2$

- b. $\sigma_{\theta_1}(E_1 \bowtie_{\theta_2} E_2) = E_1 \bowtie_{\theta_1 \wedge \theta_2} E_2$



Equivalence Rules (Cont.)

5. Theta-join operations (and natural joins) are commutative.

$$E_1 \bowtie_{\theta} E_2 = E_2 \bowtie_{\theta} E_1$$

6. (a) Natural join operations are associative:

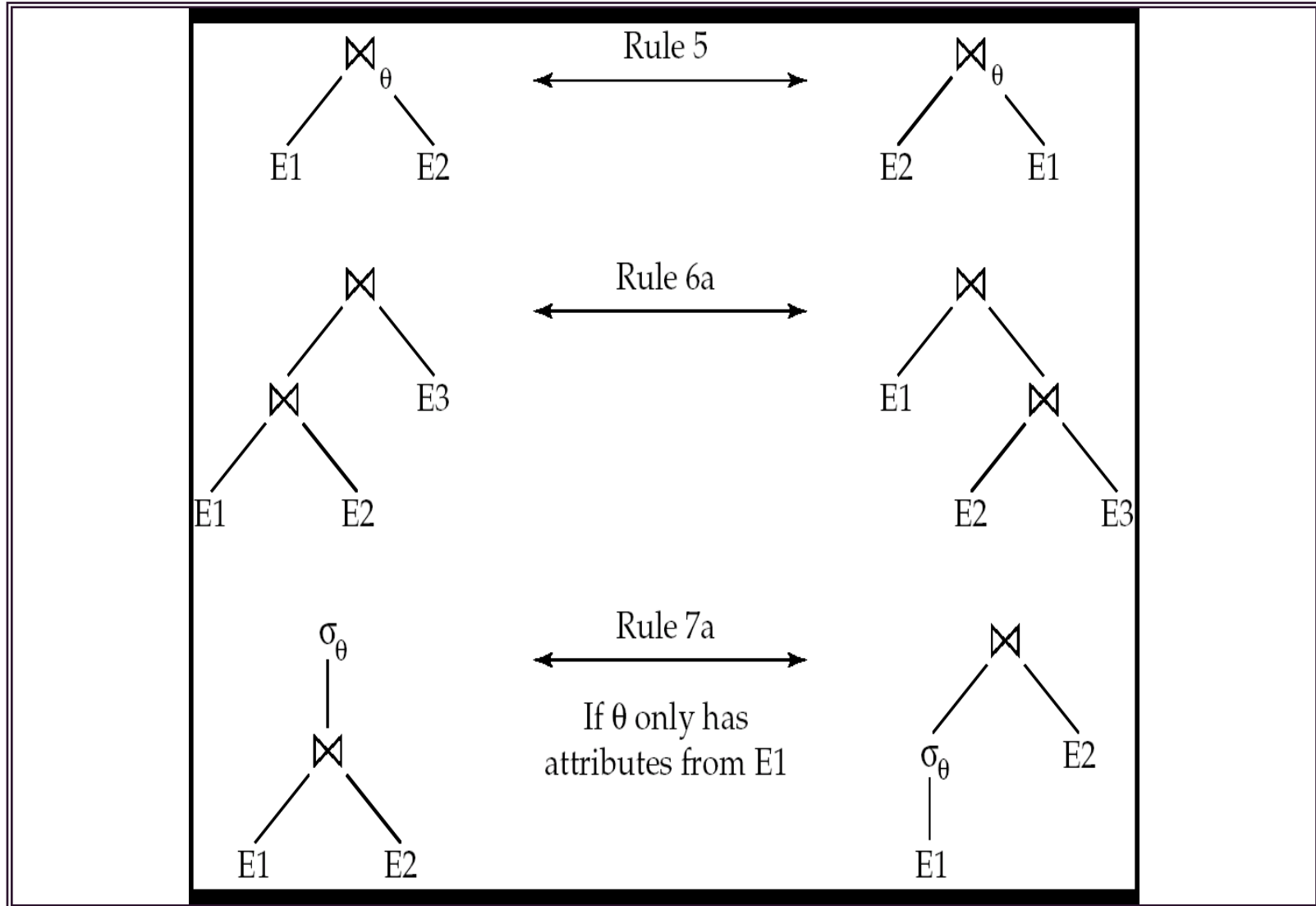
$$(E_1 \bowtie E_2) \bowtie E_3 = E_1 \bowtie (E_2 \bowtie E_3)$$

- (b) Theta joins are associative in the following manner:

$$(E_1 \bowtie_{\theta_1} E_2) \bowtie_{\theta_2 \wedge \theta_3} E_3 = E_1 \bowtie_{\theta_1 \wedge \theta_3} (E_2 \bowtie_{\theta_2} E_3)$$

where θ_2 involves attributes from only E_2 and E_3 .

+ Pictorial Depiction of Equivalence Rules





Equivalence Rules (Cont.)

7. The selection operation distributes over the theta join operation under the following two conditions:

- (a) When all the attributes in θ_0 involve only the attributes of one of the expressions (E_1) being joined.

$$\sigma_{\theta_0}(E_1 \bowtie_{\theta} E_2) = (\sigma_{\theta_0}(E_1)) \bowtie_{\theta} E_2$$

- (b) When θ_1 involves only the attributes of E_1 and θ_2 involves only the attributes of E_2 .

$$\sigma_{\theta_1 \wedge \theta_2}(E_1 \bowtie_{\theta} E_2) = (\sigma_{\theta_1}(E_1)) \bowtie_{\theta} (\sigma_{\theta_2}(E_2))$$

+ Equivalence Rules (Cont.)

8. The projections operation distributes over the theta join operation as follows:

(a) if Π involves only attributes from $L_1 \cup L_2$:

$$\Pi_{L_1 \cup L_2}(E_1 \bowtie_{\theta} E_2) = (\Pi_{L_1}(E_1)) \bowtie_{\theta} (\Pi_{L_2}(E_2))$$

(b) Consider a join $E_1 \bowtie_{\theta} E_2$.

- Let L_1 and L_2 be sets of attributes from E_1 and E_2 , respectively.
- Let L_3 be attributes of E_1 that are involved in join condition θ , but are not in $L_1 \cup L_2$, and
- let L_4 be attributes of E_2 that are involved in join condition θ , but are not in $L_1 \cup L_2$.

$$\Pi_{L_1 \cup L_2}(E_1 \bowtie_{\theta} E_2) = \Pi_{L_1 \cup L_2}((\Pi_{L_1 \cup L_3}(E_1)) \bowtie_{\theta} (\Pi_{L_2 \cup L_4}(E_2)))$$



Equivalence Rules (Cont.)

9. The set operations union and intersection are commutative

$$E_1 \cup E_2 = E_2 \cup E_1$$

$$E_1 \cap E_2 = E_2 \cap E_1$$

□ (set difference is not commutative).

10. Set union and intersection are associative.

$$(E_1 \cup E_2) \cup E_3 = E_1 \cup (E_2 \cup E_3)$$

$$(E_1 \cap E_2) \cap E_3 = E_1 \cap (E_2 \cap E_3)$$

11. The selection operation distributes over $\cup$, $\cap$ and $-$.

$$\sigma_{\theta}(E_1 - E_2) = \sigma_{\theta}(E_1) - \sigma_{\theta}(E_2)$$

and similarly for $\cup$ and $\cap$ in place of $-$

Also:
$$\sigma_{\theta}(E_1 - E_2) = \sigma_{\theta}(E_1) - E_2$$

and similarly for $\cap$ in place of $-$, but not for $\cup$

12. The projection operation distributes over union

$$\Pi_L(E_1 \cup E_2) = (\Pi_L(E_1)) \cup (\Pi_L(E_2))$$

General Transformation Rules for Rational Algebra Equations

- Some transformation rules useful in query optimization
 - Cascade of σ
 - Conjunctive selection condition can be broken up into a cascade (sequence) of individual σ operations
 - Commutativity of σ
 - Cascade of π
 - In a sequence of π operations, all but the last one can be ignored
 - Commuting σ with π

Summary of Heuristics for Algebraic Optimization

- Apply first the operations that reduce the size of intermediate results
 - Perform SELECT and PROJECT operations as early as possible to reduce the number of tuples and attributes
 - The SELECT and JOIN operations that are most restrictive should be executed before other similar operations

19.11 Summary

- Query trees
- Heuristic approaches used to improve efficiency of query execution
- Reorganization of query trees